

Member ID: _____

Time: _____

Rank: _____



LINUX OPERATING SYSTEM FUNDAMENTALS (350)

STATE 2022

CONCEPT KNOWLEDGE:

Multiple Choice (50 @ 2 points each) _____ (100 points)

TOTAL POINTS _____ (100 points)

Test Time: 60 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

MULTIPLE CHOICE

Identify the letter of the choice that *best* completes the statement or answers the question.

1. Which utility below *cannot* be used to disk manage partitions in Ubuntu?
 - a. fdisk
 - b. parted
 - c. cgdisk
 - d. gfdisk
2. What is the default partition when none is specified with the mkfs command in Ubuntu?
 - a. Ext2
 - b. Ext3
 - c. Ext4
 - d. Xfs
3. Which of the following will *not* list the disks partitions?
 - a. fdisk -l /dev/sda
 - b. gdisk -l /dev/sda
 - c. cgdisk -l /dev/sda
 - d. parted -l /dev/sda

4. Looking at the output below which are *not* directories?

```
drwxrwxr-x  2 ryan ryan 4.0K Aug 14 19:57 test
drwxrwxr-x  2 ryan ryan 4.0K Aug 14 20:06 .test1
-rw-rw-r--  1 ryan ryan   0 Aug 14 19:57 .test2
-rwxrwxr-t  1 ryan ryan   0 Aug 14 19:58 test3
```

- a. test
 - b. test1
 - c. test2
 - d. they all are
5. Which command below will persistently change the hostname to BPA1 in Ubuntu?
 - a. sudo systemctl set-hostname BPA1
 - b. sudo hostnamectl set-hostname BPA1
 - c. sudo cat BPA1 > /etc/hostname
 - d. sudo hostname --set BPA1
6. Which argument below will list all in order newest to oldest?
 - a. -t
 - b. -s
 - c. -T
 - d. -n

7. Which chmod command below will *not* make the file setup.sh executable?
 - a. chmod +X ./setup.sh
 - b. chmod +x ./setup.sh
 - c. chmod 777 ./setup.sh
 - d. they are all correct

8. The command “*sudo mount /dev/sda2 /Data*” will successfully mount the first partition of the second disk to the /Data directory.
 - a. True
 - b. False

9. Which command below will successfully list all logical volumes in the volume group named Shared?
 - a. sudo lvm Shared
 - b. sudo lvs Shared
 - c. sudo pvs Shared
 - d. sudo vgs -l Shared

10. There is an LVM Volume group named Data on this system and you’ve been asked to create a new logical volume in that volume group named Shared. That logical volume needs to be 4GB in size. Which command below will successfully complete that task?
 - a. sudo lvcreate -n Shared Project --size=4GB
 - b. sudo lvcreate -n Shared --vg Project --size=4GB
 - c. sudo lvcreate -n Shared Project -L4GB
 - d. sudo lvcreate -n Shared --vg Project -L4GB

11. Which command below will display only the ipv4 address of all network adapters on the system?
 - a. Ip -4 link
 - b. ip -4 a
 - c. ip -v4 a
 - d. ip -v4 --all

12. Which command below will successfully change the IP Address of the Network Adapter ens192 to 192.168.5.5 with subnet of 255.255.255.0?
 - a. sudo nmcli connection modify ens192 ipv4.addresses 192.168.5.5/255.255.255.0
 - b. sudo nmcli connection modify ens192 ipv4.addresses 192.168.5.5/24
 - c. sudo nmcli connection add ens192 ipv4.addresses 192.168.5.5/255.255.255.0
 - d. sudo nmcli connection add ens192 ipv4.addresses 192.168.5.5/24

13. A partition spanning the full disk has been created on disk sdc for use with LVM. What command below would correctly create a volume group named Data using that partition in Ubuntu?
- a. `sudo vgcreate -n Data -d /dev/sdc1`
 - b. `sudo vgcreate Data /dev/sdc1`
 - c. `sudo vgcreate -n Data -p /dev/sdc1`
 - d. `sudo vgcreate Data sdc1`
14. Which utility below transfers files from one server to another over ssh protocol?
- a. `tnp`
 - b. `rsync`
 - c. `tar`
 - d. `ftp`
15. Selinux is the default firewall used in Redhat Linux.
- a. True
 - b. False
16. Which command below would correctly set the NIC ens192 to use DHCP assigned IP instead of static?
- a. `sudo nmcli connection edit ens192 ipv4.method auto`
 - b. `sudo nmcli connection modify ens192 ipv4.method dhcp`
 - c. `sudo nmcli connection edit ens192 ipv4.method dhcp`
 - d. `sudo nmcli connection modify ens192 ipv4.method auto`
17. Which command below would correctly change the GID of a user named John to 1006?
- a. `sudo usermod -G 1006 Tom`
 - b. `sudo usermod -g 1006 Tom`
 - c. `sudo useradd -g 1006 Tom`
 - d. `sudo useradd -G 1006 Tom`
18. What command below would correctly create a new user called Larry with a UID of 1003?
- a. `sudo useradd -U 1003 Larry`
 - b. `sudo usermod -u 1003 Larry`
 - c. `sudo useradd -u 1003 Larry`
 - d. `sudo usermod -U 1003 Larry`
19. On the server is a user named Aaron and he needs to be added to the two groups, Sales and Project without affecting his primary group. Which command below would successfully accomplish this?
- a. `sudo usermod -aG Sales,Project Aaron`
 - b. `sudo usermod -g Sales,Project Aaron`
 - c. `sudo usermod -ag Sales,Project Aaron`
 - d. `sudo usermod -G Sales,Project Aaron`

20. You need to add a new user named Lucy but they need their home folder to be /mnt/Users/Lucy instead of in the default /home directory, which command below would successfully accomplish this?
- a. `sudo useradd -h /mnt/Users/Jane Lucy`
 - b. `sudo useradd -m /mnt/Users/Jane Lucy`
 - c. `sudo useradd -d /mnt/Users/Jane Lucy`
 - d. `sudo useradd -s /mnt/Users/Jane Lucy`
21. The command “pvcreate” is used to format a disk or partition for use with LVM volume groups.
- a. True
 - b. False
22. The Linux command ‘lsblk’ will list all disk partitions and their file system type?
- a. True
 - b. False
23. Any user added to the root group will be able run use the sudo command in Ubuntu.
- a. True
 - b. False
24. What command below will update the available kernel packages in Ubuntu?
- a. `sudo apt update`
 - b. `sudo dpkg --update`
 - c. `sudo apt upgrade`
 - d. `sudo apt dist-upgrade`
25. Ubuntu Server minimal install does not include any GUI desktop.
- a. True
 - b. False
26. The Ubuntu command ‘apt search http’ would search repositories for any packages matching the name http.
- a. True
 - b. False
27. Which command below can be used to upgrade the currently installed Ubuntu to the latest distribution version?
- a. `sudo apt dist-upgrade`
 - b. `sudo apt upgrade`
 - c. `sudo do-release-upgrade`
 - d. `sudo apt full-upgrade`

28. In Ubuntu which command would start the OpenSSH service and set it to run at boot?
- a. `sudo systemctl start sshd`
 - b. `sudo systemctl enable sshd`
 - c. `sudo systemctl enable --now sshd`
 - d. `sudo systemctl is-enabled --true sshd`
29. Which command would stop the Samba service from starting on the next boot in Ubuntu?
- a. `sudo systemctl stop samba`
 - b. `sudo systemctl disable smb`
 - c. `sudo systemctl stop --disable smb`
 - d. `sudo systemctl is-disabled --now samba`
30. Using the journalctl utility to view system logs what command below would allow you to filter the logs for anything referencing the httpd service?
- a. `sudo journalctl --search httpd`
 - b. `sudo journalctl -u httpd`
 - c. `sudo journalctl -t httpd`
 - d. `sudo journalctl -S httpd`
31. Which command below will filter journalctl output showing only warning and above type entries?
- a. `sudo journalctl -s warning`
 - b. `sudo journalctl -f warning`
 - c. `sudo journalctl -p warning`
 - d. `sudo journalctl -b warning`
32. Looking at the attributes below for the file labeled “file1” you can conclude that the user ryan would have both read and write access to the file.
- ```
-rw-r--rw- 1 ryan admins 0 Aug 28 18:08 file1
```
- a. True
  - b. False
33. Looking at the attributes below for the file labeled “file2” which user would have both read and execute access to the file?
- ```
-r-xrw---x 1 ryan zack   0 Aug 28 18:11 file2
```
- a. ryan
 - b. zack
 - c. Everyone
 - d. Neither

34. You have a user that says they can't open up a folder called /Project and get permission denied when trying to run `cd /Project`. You have already confirmed their group has been assigned to the folder. What directory permission do they need added for their group in order to open the folder?
- Write
 - Read
 - Execute
 - All of these are correct
35. If you were to change permissions to a file using the command: "`chmod 764 filename`" what would the resulting permissions look like when viewing the file?
- `-rwxrwx-r--`
 - `-rwxr-----`
 - `-rw--wxr--`
 - `----r--rwx`
36. Review the permissions below for the folder Database. You've been asked to change permissions so that only the root user and members of the Management group have read/write access.
- ```
drwxr-xrwx 2 root Management 4.0K Aug 28 18:34 Database
```
- `sudo chown 707 Database`
  - `sudo chmod 707 Database`
  - `sudo chmod 770 Database`
  - `sudo chown 770 Database`
37. If you were to set permissions of 646 on a script called `runme.sh` users in which category would be given the ability to execute the script?
- Owner
  - Group
  - Others
  - None of these are correct
38. The `setfacl` command allows you to set permissions for users and groups on a folder or file in Ubuntu.
- True
  - False
39. Which command below would successfully filter out any mention of errors from the `/var/log/syslog` file and output them to a file called `/var/log/errors.log` overwriting any existing data?
- `grep -i "error" /var/log/syslog > /var/log/errors.log`
  - `find /var/log/syslog -name "error" >> /var/log/errors.log`
  - `grep -i "error" /var/log/syslog >> /var/log/errors.log`
  - `find /var/log/syslog -name "error" > /var/log/errors.log`

40. The SSH service uses port 21 by default.
- True
  - False
41. Which port would you need to enable for https on Linux?
- 80
  - 445
  - 443
  - 465
42. When viewing processes with the top utility which key would sort the list by Memory usage?
- T
  - C
  - P
  - M
43. Which command below can be used to only list process usage by a specific user named Erin?
- top -u Erin
  - top -S Erin
  - top -U Erin
  - top -s Erin
44. You need to add a new disk partition, *sdb1*, into your existing volume group named *VOL1*. Which command below will accomplish this?
- sudo vgreszie VOL1 /dev/sdb1
  - sudo vgetend VOL1 /dev/sdb1
  - sudo vgexpand VOL1 /dev/sdb1
  - sudo vgadd VOL1 /dev/sdb1
45. All data has been migrated off of disk *sdb1* and is now ready to be removed from the volume group *Database*. What command below will successfully complete the removal?
- vgremove Database /dev/sdb1
  - vgreduce Database /dev/sdb1
  - pvremove Database /dev/sdb1
  - pvmove Database /dev/sdb1
46. The command *pvmove* is used to migrate data off of a physical disk before it can be removed from a volume group.
- True
  - False

47. Using the Linux utility nmcli you can modify network settings and also set NICs to start up at boot.
- a. True
  - b. False
48. Which command below would be used to update grub config on Redhat Linux?
- a. grub-install
  - b. update-grub
  - c. grub-mkconfig
  - d. save-grub
49. In Redhat Linux selinux is enabled by default.
- a. True
  - b. False
50. The NFS service provides file and print services for both Linux and Windows client systems.
- a. True
  - b. False